

Let There Be Light: Learning Lights from In-The-Wild Character Images

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Figure 1: Let There Be Light: Learning Lights from In-The-Wild Character Images

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1 Overview

We present “Let There Be Light”, a real-time system that extracts discrete, production-ready lighting parameters from in-the-wild character images for direct use in tools such as Blender and Unreal Engine. Unlike traditional inverse rendering or environment map estimation, our generative neural network predicts controllable light configurations, including light types, positions, colors, and intensities, optimized for real-world animation workflows. Built on a cascade neural architecture, the system first encodes intrinsic visual cues and then decodes them into a compact, discrete lighting representation. Trained on a large curated dataset of synthetic and real imagery with ground-truth lighting setups, it achieves near real-time inference while maintaining production-level accuracy.

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On stage, we will demonstrate the system as a Blender Add-on, showcasing its robustness across diverse inputs and its ability to accelerate animation lighting design from hours to seconds. The tool empowers artists to replicate, adapt, and iterate lighting setups

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interactively, transforming what was once a labor-intensive process into an intuitive, generative experience. Beyond character lighting, the approach opens new possibilities for real-time scene authoring and cinematic lighting design in virtual production.